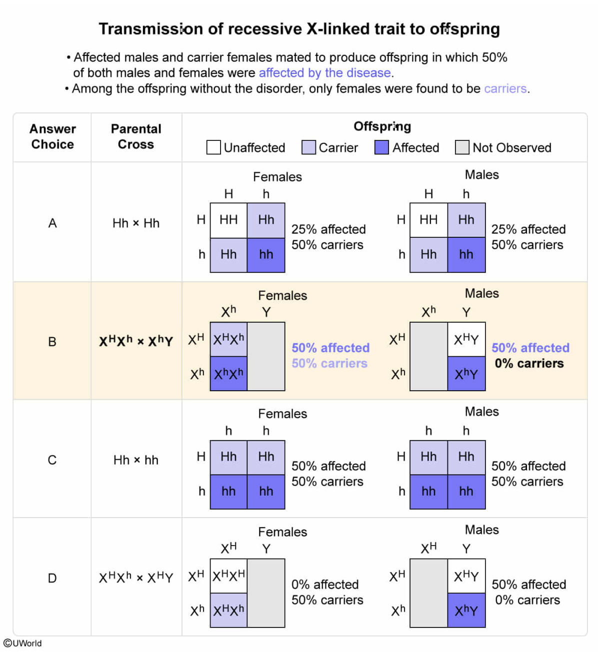
The inheritance pattern of a disease was determined in a study of affected families. Researchers found that when affected males and carrier females mated, they produced offspring in which 50% of both males and females exhibited the disease. However, among unaffected offspring, only females were found to be carriers. Which of the following best describes the parental genotypes in such a mating scenario?

- ◯ A. The mother's genotype is Hh; the father's genotype is Hh.
- ✔ ◉ B. The mother's genotype is X^HX^h; the father's genotype is X^hY. [67%]
- ◯ C. The mother's genotype is Hh; the father's genotype is hh. [3%]
- ◯ D. The mother's genotype is X^HX^h; the father's genotype is X^HY. [28%]

Correct
66% answered correctly

Explanation:



In **sex-linked inheritance**, traits arise from genes on sex chromosomes (eg, X and Y chromosomes). Females inherit two X chromosomes, whereas males inherit one X and one Y chromosome. Therefore, females (XX) and males (XY) exhibit different inheritance patterns for X-linked traits. Recessive X-linked traits are exhibited in males more often than in females because expression of recessive alleles on a male's X chromosome cannot be masked by alleles on the Y chromosome. In contrast, for females to exhibit recessive X-linked traits, the female must inherit the recessive allele on *both* X chromosomes.

In this question, researchers investigated inheritance of a disease among offspring of affected male and carrier female parents. The study found that 50% of both male and female offspring exhibited the disease; additionally, all carrier offspring were female. The question asks which choice best describes the genotypes of the parents.

Transmission of the disease in question as a recessive autosomal or recessive sex-linked disorder would be expected to result in roughly half of both male and female offspring being affected by the disorder. However, only transmission as a sex-linked recessive disorder would produce carrier offspring that were all female. Therefore, the **mother's genotype** is likely $\mathbf{X^H X^h}$, while the **father's genotype** is likely $\mathbf{X^h Y}$ (where H and h refer to the dominant and recessive alleles, respectively).

(Choices A and C) These parental genotypes would describe traits for which alleles are found on an autosome (*not* a sex chromosome). The pattern of inheritance of traits found on autosomes is typically *very similar* among male and female offspring.

(Choice D) These parental genotypes describe a mating between females carriers and affected (*not* unaffected) males.

Educational objective:
The inheritance and expression of sex-linked genes (ie, those located on the X or Y chromosome) differs between males and females. Because males have a single X chromosome, sex-linked recessive traits tend to be expressed at higher rates in males than in females.

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